

Introduction: Josh Talks to You on Paper!

Year 11 Profile Course with Ms. Fernandez and Josh

November 15 2017

Dear Student,

For the next few weeks we will be lucky enough to study functions. Now before you roll your eyes and start praying that this is one of those boring dreams where you at least wake up and it's a weekend, you should know that the concept of functions is one of the most fundamental and powerful ideas not only in mathematics, but pretty much any subject in the modern world. When we say that a person is a “product of their environment,” or that something occurred due to “unforeseen developments,” or we are reaching a “paradigm shift” in a given field or section of society, we are using elements of what one might call “functional thinking.” There is one topic in particular that I find perfectly captures many of the key attributes of functional thinking and also happens to be relevant to almost every part of our lives: Cryptography. Now, of course, we can think of cryptography and its highly related field of Information Theory as mostly the stuff of secret agents with tiny secret code notebooks, but it's way cooler than that: if what we are really talking about is transforming information from one form to another so that something important is preserved (like meaning, or colors and lines of an image, or the wave form that comprises a song) then codes and cryptography are everywhere! They are to be found in everyday actions: from we translate between languages to how we stream movies online or how we send messages securely on our phones—but also extreme cases like how we communicate with the mars rovers or how we detect earthquakes. We won't specifically cover most of those cases in class (though we could deal with simplified versions of them if you want). But after every lesson I will be giving you a set of notes and ideas (both in paper form and digitally) and the digital form will include links to YouTube videos, Khan Academy lessons, book recommendations, and other resources that you can look up on your own.

As you have already seen, there are few things we will be doing a bit differently than normal. This is a course you all technically chose to be in, so we have a bit more flexibility as to how we can design and implement it. (And it is only because of the kindness and flexibility of Ms. Fernandez that I get to do this stuff *at all* with you all, so we all owe her big time!) Let's go through a couple of the major differences that will be present for at least the next few weeks.

Problems and Problem Solving

A significant difference will probably cause the most emotional and intellectual discomfort and frustration (also known as brain-pain): that is, how problems and questions will be phrased. Here is an example of a standard problem having to do with functions:

Given that $f(x) = 3x^2 - 4$ what is the inverse function $f^{-1}(x)$?
What are the range and domain of this function? is this function injective or surjective?

Regardless of whether all of the words in the above question are known to you, I think we can all agree that this looks like a standard math question, where there is one clear answer. It is important for you all to be able to mechanically do (almost without thinking) such problems by the end of this month. Why? Well, for a few reasons, but the biggest one I can think of for this particular type of problem is that it will come up a lot in future exams, but the idea of inverses is super powerful and interesting! (the other words are pretty basic, the sound fancy but we will mostly use them as tests of our resolve: can we read such words without our eyes glazing over or without finding ourselves in a mild state of panic?) In contrast, today we will see problems like this:

A *palindrome* is a word that is the same backwards and forwards (like mom, kayak, racecar, or deified), are palindromes still palindromes after we encrypt them with a shift cipher? Can you prove it? What if we use some other type of cipher on the same alphabet? What notation might you use to prove it with fewer words or symbols? (Be sure to clearly define what you mean by each symbol or abbreviation!)

So the first thing we might notice is that there are more words in this question, and that it involves proving something, or arguing something. In this case the argument is not likely to be very long, but already you are being invited to think of the problem differently—what would you need to do this problem? There is still a “correct” answer, but there are many ways of approaching it and writing it. The key thing that we expect here is not just that you have the write answer, but that you are able to write it in such a way that you convince someone else. I promise to make the initial problems such that at least the key elements can be explained in a short paragraph, and once we have a common notation things will roll along much more smoothly, but this work will hopefully cause your brain to sweat—meaning that you will have to think, as opposed to regurgitate facts or “remember” how to do things. One needs to do that stuff in life too, but here we have the chance to spread our wings a bit.

How We Work: Flexible with Feedback

I will attempt to do as little lecturing as possible over the next month. Sometimes, like when we decide on notation or if the same issues keep coming up,

me or Ms. Fernandez getting up and saying things or writing stuff on the chalk board is just the best solution. Except for those rare occasions you peeps are on your own! You can choose to work in groups, in partners or individually or some combination thereof. It is expected that if there is an issue or question you ask your peers before asking me or Ms. Fernandez. All work you do should have the names of people you worked with, we highly recommend trying problems on your own first, and then moving to partners or small groups (we will model this during our first lesson, but after that it is up to you).

At the end of every lesson we will take 15 minutes to write a description of what you did that day along with any solutions or questions you had, things you learned and concepts you think you understand or still haven't quite grokked. If you had a rough morning and really didn't get much done, or if you found the material boring, or perhaps you got really confused by something Schlomo said¹ then you can always write about what happened there! How did you figure it out in the end? Are you still stuck? Are my jokes getting to old? Are you just generally frustrated? Are you having some relationship problems? (okay that last one we can't really help with...) You get the idea!

Not only does this give Ms. Fernandez and I feedback about what happened today and how things went and what we can do better, but at the end you will have a record of what it was like to learn a new thing: something we rarely get in life—you don't really have a record of the time it took you to learn to speak, or to write, or to read, these are just things you can do now! The goal of this is to make this process of learning a bit more transparent, but also allow you to do things at your own pace. Ms. Fernandez and I will have a next to endless amount of questions and follow up questions we can ask you, but the good part is you can also just make up your own—so long as it has to do with functions and you are learning something in the process and (super important!) you are able to *explain this relationship* to what we are learning (which is a whole different level of thinking), we are happy.

Why We are Doing This to You: Building Creativity and Flexibility

The goal here, believe it or not, is for this style of lesson to be more engaging and encourage more creativity and mathematical play: in other words, this is suppose to be fun. If it isn't fun, if you all don't buy into this stuff at all, then there is always the standard lecture and problems method—which we can do if you really want. These methods are ones you can definitely take advantage of, you could sit around and not really work and come out of this month without much understanding of how functions work and how the various topics (like cryptography and encoding and algorithms and gnomons and figurate numbers and recursive functions on graphs) all fit together. But this method, where you

¹Schlomo is a character who will be very important to all future classes, as we will blame most things that go wrong on him and use him for a variety of experiments and examples, at times with his equally incompetent sister Schlomi. His grandparents Bubie and Zadie might also appear in examples as their names do not share first letters.

guys have more control over how you work, means you will get as much out of it as you put in. If you put in the time and mental/emotional effort, we will match you with 120%, but if you sit there and claim boredom and refuse our offers to help or give you other problems, the missed opportunity is yours. On the more positive side, I am super excited to try out this method with you, as it more closely aligns to how I think math class (and other classes) might be better presented. This can only work as Ms. Fernandez is already pretty open and trusting of you all, but for the next month, we get to take this classroom level openness, trust and freedom to a whole new level and learn some really fascinating stuff along the way. The problems we will solve and the questions we will tackle together are ones that often remained unsolved for millennia—but the coolest part is that we get to solve them anew! We get to share in the joy of the first person who cracked these old chestnuts! Thousands of problem solvers sought to use these and other mathematical tools to shape how they saw the world but also literally to shape the world itself. This will be a journey where we truly create mathematics together: I am looking forward to embarking on it with you all and seeing where we end up!

Josh

Joshua S. Aaron

Summary

Schedule

- First 15 Minutes: Review and questions, distribution of new material, announcements.
- 15 minutes to 45 minutes: Work on material at your own pace.
- 45 minutes to 50 minutes: Break
- 50 minutes to 60 minutes: Questions, clarifications, resettle
- 60 minutes to 80 minutes: Work on material at your own pace
- 80 minutes to 95 minutes: Write daily summary, questions, struggles, successes, concerns, commit to work for the week, submit to the old people.
- 95 minutes to next week: Get on with your life, ask questions via email if you have pressing concerns, general questions, or want support, resources.

Principles

-
-
-
-